

SQL Fundamentals.

Selecting and Sorting

SQL Query

① SELECT *
FROM employees table ;

Expected output

RESULTS	id	first_name	last_name	department	salary	hire_date	city
	1	Alice	Green	IT	70000	2020-01-10	Johannesburg
	2	Brian	Lee	HR	45000	2019-03-22	Cape Town
	3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
	4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
	5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

② SELECT DISTINCT (department)
FROM employees table ;

Expected output

RESULTS	department
	IT
	HR
	Finance
	Marketing

③ SELECT first_name,
last_name
FROM employees table
WHERE ORDER By 'Salary' DESC;

Expected output

RESULTS	first_name	last_name
	Eva	Naidoo
	Alice	Green
	Cathy	Zulu
	David	Mokoena
	Brian	Lee

④ SELECT id,
first_name,
last_name,
salary
FROM employees table
WHERE ORDER BY salary DESC
LIMIT 3;

Expected output

RESULTS	id	first_name	last_name	salary
	5	Eva	Naidoo	72000
	1	Alice	Green	70000
	3	Cathy	Zulu	65000

Filtering with WHERE

(5)
 SELECT id,
 first_name,
 last_name,
 department
 FROM employees table
 WHERE department = 'IT';

Expected output

RESULTS	id	first_name	last_name	department
	1	Alice	Green	IT
	5	Eva	Naidoo	IT

(6)
 SELECT id,
 first_name,
 last_name,
 department,
 salary
 FROM employees table
 WHERE department = 'Finance' AND salary > 60000;

↳ Expected output

RESULT	id	first_name	last_name	department	salary
	3	Cathy	Zulu	Finance	65000

(7)
 SELECT id,
 first_name,
 last_name,
 department

RESULT	id	first_name	last_name	department
	2	Brian	Lee	HR
	4	David	Mokoena	Marketing

FROM employees table

WHERE department = 'HR' OR department = 'MARKETING';

8 SELECT id,
first_name,
last_name,
department
FROM employees table
WHERE NOT department = 'IT';
Expected output ↓

RESULTS	id	first_name	last_name	department
	2	Brian	Lee	HR
	3	Cathy	Zulu	Finance
	4	David	Mokoena	Marketing

9 SELECT id, first_name, last_name, department
FROM employees table
WHERE 'employee' IN ('IT', 'HR', 'Finance');

RESULTS	id	first_name	last_name	department
	5	Eva	Naidoo	IT
	1	Alice	Green	IT
	2	Brian	Lee	HR
	3	Cathy	Zulu	Finance

10 SELECT id,
first_name,
last_name,
department,
Salary,
City
FROM employees table
WHERE 'employee' IN ('IT') AND Salary > 65000 AND
city ('Johannesburg').

Expected output ↓

RESULTS	id	first_name	last_name	department	salary	city
	5	Eva	Naidoo	IT	72000	Johannesburg
	1	Alice	Green	IT	70000	Johannesburg